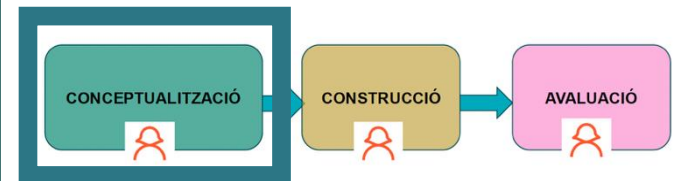




Generada
amb DALLÉ

TEMA 2 FHiC

Conceptualització del disseny (Part 2)



 UNIVERSITAT DE
BARCELONA

Inmaculada Rodríguez Santiago
Facultat de Matemàtiques i Informàtica
Curs 25-26

"We don't design for ourselves, we design for our users. User research is the tool that allows us to understand them."

Jakob Nielsen

CONTINGUT

PART 2

(2.3, 2.4, 2.5)

2.1 Introducció

2.2 Tècniques d'investigació d'usuaris

2.3 Models i artefactes

2.4 Investigació d'usuaris en productes amb IA

2.5 IAG en la Conceptualització

CONTINGUT

2.1 Introducció

2.2 Tècniques d'investigació d'usuaris

2.3 Models i artefactes

2.4 Investigació d'usuaris en productes amb IA

2.5 IAG en la Conceptualització

2.3 Models i artefactes

2.3.1 Introducció

2.3.2 Grups d'usuaris

2.3.3 Persona

2.3.4 Empathy map

2.3.5 Scenarios

2.3.6 User Journey Map



2.3 Models i artefactes

2.3.1 Introducció

La recerca d'usuaris permet crear **models** que representen la seva realitat. Aquests models es concreten en **artefactes** que ajuden els dissenyadors a planificar, comunicar i crear dissenys centrats en l'usuari. És a dir, serveixen per:

- Empatitzar amb l'usuari.
- Definir o redefinir el problema a resoldre.
- Tenir a l'usuari present al llarg de tot el procés.
- Comunicar idees dins de l'equip i amb altres agents implicats.
- Generar i orientar idees de disseny.

2.3 Models i artefactes

2.3.1 Introducció

En la conceptualització, dos aproximacions per elaborar els models i artefactes:

- Ens basem en dades recollides amb tècniques de ***user research***.
- Ens basem en **assumpcions** (info d'experts en domini i altres implicats)
 - En aquest cas, hem de re-ajustar els artefactes després de l'apàrs al *user research*.

2.3 Models i artefactes

2.3.1 Introducció

2.3.2 Grups d'usuaris

2.3.3 Persona

2.3.4 Empathy map

2.3.5 Scenarios

2.3.6 User Journey Map

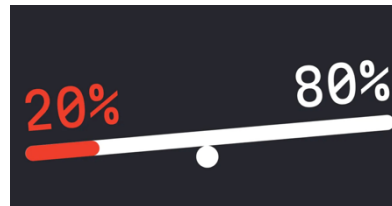


2.3 Models i artefactes

2.3.2 Grups d'usuaris

Grups d'usuaris segons la relació amb el producte:

- *Usuaris primaris*: utilitzen el producte de forma directa, són para qui dissenyem les funcionalitats principals (e.g. repartidor que utilitza una app de rutes).
- *Usuaris secundaris*: no és el públic principal per al qual s'ha dissenyat, però es veuen afectats d'alguna manera (e.g. el supervisor dels repartidors).
- *Usuaris extrems*: casos límits que amplifiquen necessitats (e.g. molt experts en tecnologia, usuaris amb diversitat funcional). Tot i la regla 80/20 de Pareto (que es focalitza en la majoria), tenir-los en compte pot beneficiar a tothom i impulsa l'innovació.



“El 80% de la terra es propietat de 20% de la població” Pareto, 1896

“El 80% d'usuaris utilitzen el 20% de les funcionalitats”

2.3 Models i artefactes

2.3.2 Grups d'usuaris

Grups d'usuaris segons característiques:

- Demogràfics (edat, nivell educatiu, context cultural...)
- Nivel d'expertesa (ex. novell, intermedis, experts)
- Conductuals (freqüència d'ús, context d'ús, ...)
- Accesibilitat (diversitat funcional visual, auditiva o motora, dificultats cognitives)
- Rol (ex. professor, estudiant)
- ...

2.3 Models i artefactes

2.3.1 Introducció

2.3.2 Grups d'usuaris

2.3.3 Persona

2.3.4 Empathy map

2.3.5 Scenarios

2.3.6 User Journey Map

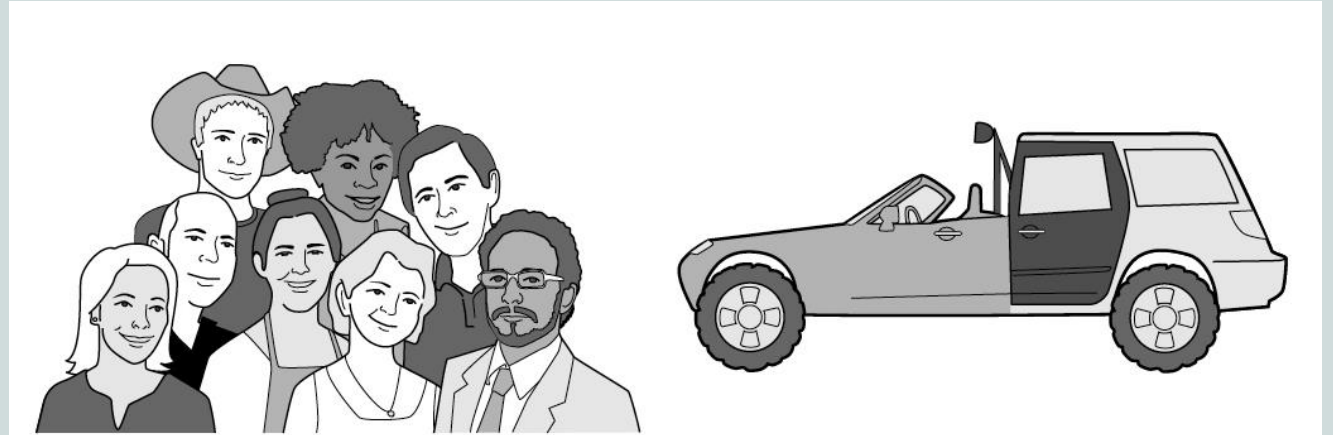


2.3 Models i artefactes

2.3.3 Persona



Font Flaticon



Alan Cooper, 2007

2.3 Models i artefactes

2.3.3 Persona



Font Flaticon

Pràctica 2

(Es defineix 1 Persona)

- OBJECTIU: Crear personatges ficticis (arquetipus d'usuaris) que representin els usuaris i donin suport al procés de disseny.
- Consisteix en crear perfils d'usuaris imaginaris basats en la informació recollida a les entrevistes, qüestionaris, observacions, etc.
- Cada Persona té característiques demogràfiques (foto, edat, gènere, gustos, hàbits,...), objectius, comportament, motivació, frustracions... sempre relacionats amb el àmbit del projecte.

<https://digital.gov/2023/05/19/personas-learn-how-to-discover-your-audience-understand-them-and-pivot-to-address-their-needs/>

2.3 Models i artefactes

2.3.3 Persona



Font Flaticon

Beneficis

- Faciliten una comprensió compartida dels grups d'usuaris.
- Donen context a les dades amb històries fàcils de recordar.
- Guien les decisions de disseny segons les necessitats de cada persona.
- Ajuden a la prioritització de funcionalitats en funció de com satisfan les necessitats d'una o més persones.
- Humanitzen les dades i fomenten l'empatia

Cooper, 1999

2.3 Models i artefactes

2.3.3 Persona



Font Flaticon

Riscos

- Una sola persona no descriu totes les característiques dels usuaris.
- Encara amb varies persones, no poden representar tota la combinació ni la diversitat de perfils existents.
- Una persona no és un usuari real.
- El disseny pot acabar centrant-se només en les persones definides.

Example

Clark Andrews

AGE 26

OCCUPATION Software Developer

STATUS Single

LOCATION San Jose, CA

TIER Experiment Hacker

ARCHETYPE The Computer Nerd

Friendly

Clever

Go-Getter



"I feel like there's a smarter way for me to transition into a healthier lifestyle."

Motivations

Incentive

Fear

Achievement

Growth

Power

Social

Goals

- To cut down on unhealthy eating and drinking habits
- To measure multiple aspects of life more scientifically
- To set goals and see and make positive impacts on his life

Frustrations

- Unfamiliar with wearable technology
- Saturated tracking market
- Manual tracking is too time consuming

Bio

Aaron is a systems software developer, a "data junkie" and for the past couple years, has been very interested in tracking aspects of his health and performance. Aaron wants to track his mood, happiness, sleep quality and how his eating and exercise habits affects his well being. Although he only drinks occasionally with friends on the weekend, he would like to cut down on alcohol intake.

Personality

Extrovert

Introvert

Sensing

Intuition

Thinking

Feeling

Judging

Perceiving

Technology

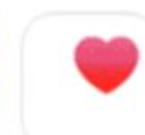
IT & Internet

Software

Mobile Apps

Social Networks

Brands



Plantilla a Figma (Pràctica 2)

Es pot
modificar
adaptar-la al
vostre
projecte.

Kristin Watson



AGE	27
EDUCATION	Masters in Business
STATUS	Single
OCCUPATION	Sales Manager
LOCATION	Sydney
TECH LITERATE	High

“ I am used to with online service and I usually do my online shopping from Instagram.

Personality

Introvert

Thinker

Spender

Tech-savy

Bio

She currently lives in Sydney. She finished her master in business and has just been promoted to Sales Manager. She is currently single and like to go out with friends on long holidays.

Core needs

- Need to find people with similar skills that can help her tackle company goals.
- View all her hirings in an overview
- The price of the service is very important

Frustrations

- Price is high related to quality they provide
- Currently finds perfect people from past work relations, family, friends and within my circle and online which is tedious
- Not much choice and comparison not available

Brands



facebook

YouTube



Payment medium



Cash/Cheque



Digital Payment

Platform



Website



Mobile App

2.3 Models i artefactes

2.3.3 Persona



Font Flaticon

Proto-persona

És una representació preliminar i provisional d'un tipus d'usuari, creada a partir de les hipòtesis, experiència i coneixement intern de l'equip, abans de disposar de dades empíriques suficients.

Anti-personas

És la representació d'un perfil d'usuari per al qual el producte no es dissenya.

2.3 Models i artefactes

2.3.3 Persona



Font Flaticon

Data-driven Persones (DDP)

- La IA (Machine Learning i Large Language Models) permeten crear Persones dinàmiques, predictives i accionables a partir de grans volums de dades.
- En lloc de crear “perfils estàtics”, es descobreixen automàticament i s’actualitzen en temps real.

<https://think.design/blog/how-to-create-personas-driven-by-data/>

<https://s2p.qcri.org/blog/turning-data-driven-profiles-into-truly-interactive-personas/>

[Using AI for User Representation: An Analysis of 83 Persona Prompts.](https://osf.io/rvnax/files/ygz9e?view_only=898e41f4bd47447583517dba87a73d02)

https://osf.io/rvnax/files/ygz9e?view_only=898e41f4bd47447583517dba87a73d02

Personas

Exercici: (Veure activitat a Material pràctiques del CV)

- Primer, crear Proto-persona (amb assumpcions i hipòtesis del l'equip de disseny, abans de conèixer de primera mà els usuaris)
- Després, definir Persona basant-nos en dades recopilats d'usuaris (5 entrevistes)
- Finalment, anti-persona, és a dir, per a qui NO dissenyem.



✓ PREREQUISITES

- ✓ DOMAIN KNOWLEDGE
- ✓ OPTIONAL: PROVISIONAL PERSONAS
- ✓ LIKELY ROLES ARE IDENTIFIED
- ✓ RESEARCH PLAN



PERSONA CREATION IS
BASED ON RESEARCH



USER INTERVIEWS—
START WITH 5/ROLE
(THEN ADD MORE LATER—
DOMAIN COMPLEXITY
IS THE KEY FACTOR HERE)

ADDITIONAL RESOURCES



CONTEXTUAL
INQUIRY



SURVEY



DIARY
STUDIES

CREATING PERSONAS

PART 1

UX Knowledge Base Sketch #13

PERSONA: REPRESENTATION OF A GROUP OF USERS,
AN ARCHETYPE, THAT REFLECTS PATTERNS BASED ON
BEHAVIOR, GOALS, ATTITUDE AND OTHER VARIABLES.



BENEFITS

- ↑ USER-CENTERED MINDSET
- ↑ HELPS DESIGN DECISIONS:
"IS THIS SOLUTION GOOD FOR
THIS PERSONA?"



ROLE ≠ PERSONA

↓ DEFINED BY ↓
TASKS BEHAVIOR, GOALS,
 NEEDS, CHALLENGES ETC.



GO THROUGH YOUR NOTES
SEARCH FOR:

- GOALS & NEEDS
- PAIN POINTS / CHALLENGES
- TASK FREQUENCY
- PRIORITIES
- MENTAL MODELS
- SKILLS
- WHAT / WHO PLAYS AN IMPORTANT
ROLE IN THE PROCESS (E.G. LIGHT
CONDITIONS OR ANOTHER PERSON)
- DEMOGRAPHICS - ONLY IF IT IS
CLOSELY RELATED TO BEHAVIOR

>>> TO BE CONTINUED IN PART 2! <<<



WHO ARE YOU
DESIGNING FOR?



HOW MANY PERSONAS
SHOULD BE CREATED?

DEPENDS ON THE COMPLEXITY,
2-6/ROLE IS GOOD
IF IT IS NOT THAT COMPLEX.



BRAINSTORMING
SESSION WITH
THE DESIGN TEAM

GOAL: FINDING PATTERNS



WRITE ~20 VARIABLES
ON A WHITEBOARD

DO 1 ROLE AT A TIME

VARIABLES CAN HAVE 3 FORMS:

SPECTRUM

SOMETHING IS
IMPORTANT

SOMETHING IS
NOT IMPORTANT

PAIR

● YES

○ NO

MULTIPLE
CHOICE

REASON FOR...

□ REASON 1 □ REASON 2

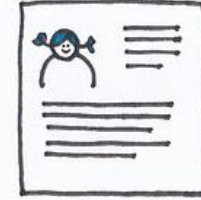
□ REASON 3 □ REASON 4



CREATING PERSONAS

UX Knowledge Base Sketch #14

PART 2



©KRISZTINA SZEROVAY
WWW.SKETCHINGFORUX.COM



- ✓ PREREQUISITES
- 🔍 RESEARCH TYPES
- 📈 DEFINITION & BENEFITS
- 📄 HOW TO GO THROUGH YOUR NOTES (SEARCH FOR...)
- 💡 BRAINSTORMING: CREATE A LIST OF VARIABLES

- 📋 LIST THE VARIABLES BY ROLES
- 3 FORMS:
 - SPECTRUM
 - PAIR
 - MULTIPLE CHOICE



AT THIS POINT YOU CAN START THE CROSS-CASE ANALYSIS

START TO ASSIGN INTERVIEWEES TO THE VARIABLES

YOU SHOULD HAVE A LIST OF BEHAVIORAL VARIABLES, E.G.:

- GOALS
- ATTITUDE
- TASK FREQUENCY
- PRIORITIES
- MENTAL MODELS

ADD DEMOGRAPHIC VARIABLES, IF IT IS RELEVANT (E.G. IF YOU ARE DESIGNING FOR KIDS OR ELDERLY USERS)

EXAMPLE: ONLINE LIBRARY
ROLE: BORROWER



- TASK FREQUENCY - SPECTRUM:

DOES NOT BORROW OFTEN ————— BORROWS OFTEN
- GOALS - MULTIPLE CHOICE

REASONS FOR BORROWING
REASON1 REASON2 REASON3
- OCCUPATION - PAIR

STUDENT NOT STUDENT

FOR EXAMPLE YOU HAVE INTERVIEWED 5 USERS (BORROWERS)

2 APPROACH:

- DO THE ASSIGNING VARIABLE BY VARIABLE ASSIGNING EACH INTERVIEWEE
- CONCENTRATE ON ONE INTERVIEWEE AT A TIME

OR

DOES NOT BORROW OFTEN ————— BORROWS OFTEN

Tara Nora Eric Anna Tom

OCCUPATION
STUDENT
Tom, Anna

NOT STUDENT
Eric, Nora, Tara

THEN SEARCH FOR PATTERNS
↓
MARK THE INTERVIEWEES WHO OFTEN APPEAR CLOSE TO EACH OTHER

⚠️ IF THERE IS A MISMATCH, THINK ABOUT WHY IS THAT

📊 BASED ON THE PATTERNS, CREATE A LIST OF CHARACTERISTICS

🎯 DEFINE THE PERSONA'S GOALS

+ ADD

- BEHAVIORS
- TASKS
- MOTIVATIONS
- PAIN POINTS
- FEELINGS

+ IF RELEVANT

- SKILLS
- ENVIRONMENT
- DEPENDENCIES



CHOOSE AN APPROPRIATE

- NAME
- PHOTO
- SOME DEMOGRAPHICS (AGE, GENDER)

🔄 DON'T FORGET TO ITERATE

📢 COMMUNICATE THE PERSONAS TO THE TEAM

2.3 Models i artefactes

2.3.1 Introducció

2.3.2 Grups d'usuaris

2.3.3 Persona

2.3.4 Empathy map

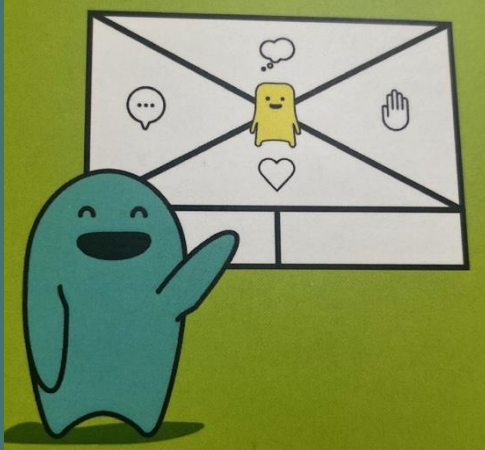
2.3.5 Scenarios

2.3.6 User Journey Map



2.3 Models i artefactes

2.3.4 Empathy map



Font www.designthinking.es

OBJECTIU: Crear una representació visual de què pensa, què fa, què sent, què escolta, què veu l'usuari en un context específic.

Ajuden a veure el producte o servei des de una perspectiva de l'usuari.

- Es crea un Empathy per cada Persona definida.
- N'hi ha diversos formats però tots amb la mateixa essència.



MAPPING THE USERS' THOUGHTS,
NEEDS, MOTIVATIONS, FEELINGS
ATTITUDES, BELIEFS, ISSUES



- GREAT TEAM EXERCISE
- VISUALIZE USERS' NEEDS
- GOOD STARTING POINT
- QUICK & COST-EFFICIENT



THERE ARE
DIFFERENT
TEMPLATES

OTHER ASPECTS /WORDING
FOR EXAMPLE:

- TASKS
- INFLUENCES
- QUOTES
- GOALS



PREREQUISITES:

- UX RESEARCH DATA
- PERSONAS



DON'T FORGET TO
DOCUMENT THE SESSION!

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WWW.SKETCHINGFORUX.COM

EMPATHY MAP

UX Knowledge Base Sketch #12



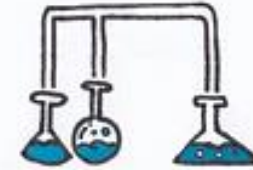
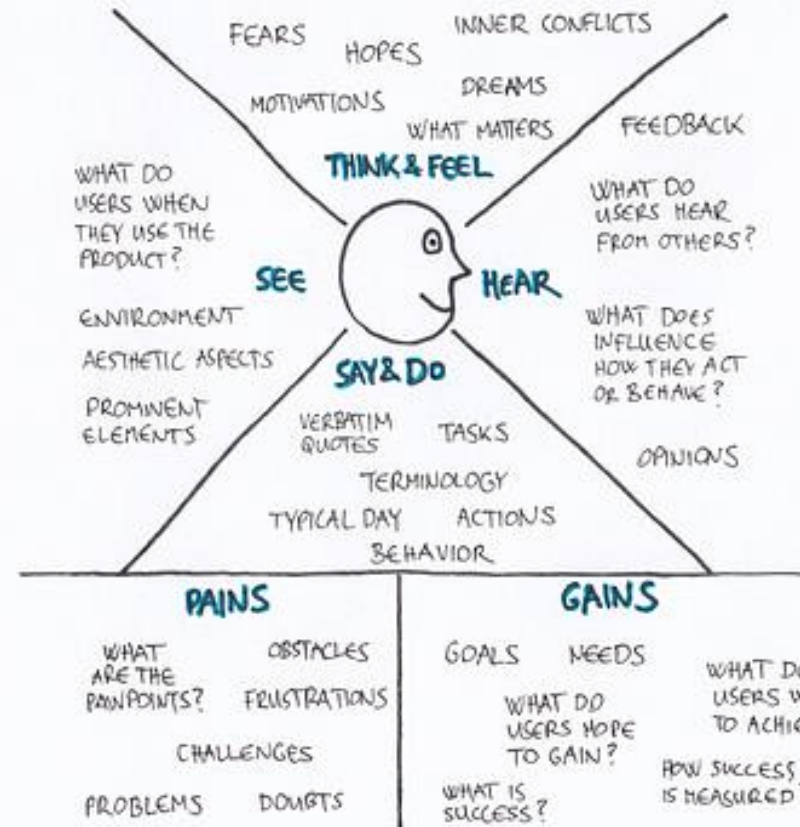
CREATE A MAP FOR
EACH PERSONA
(/CUSTOMER SEGMENT)



AFTER
COLLECTING
RESEARCH DATA



BEFORE DEFINING
THE PRODUCT REQUIREMENTS



WAY TO SYNTHESIZE
RESEARCH DATA TO BETTER
UNDERSTAND THE USERS



PARTICIPANTS
OF THE MAPPING

↳ CORE TEAM MEMBERS

↳ OPTIONAL: STAKEHOLDERS



TOOLS FOR THE
MAPPING SESSION

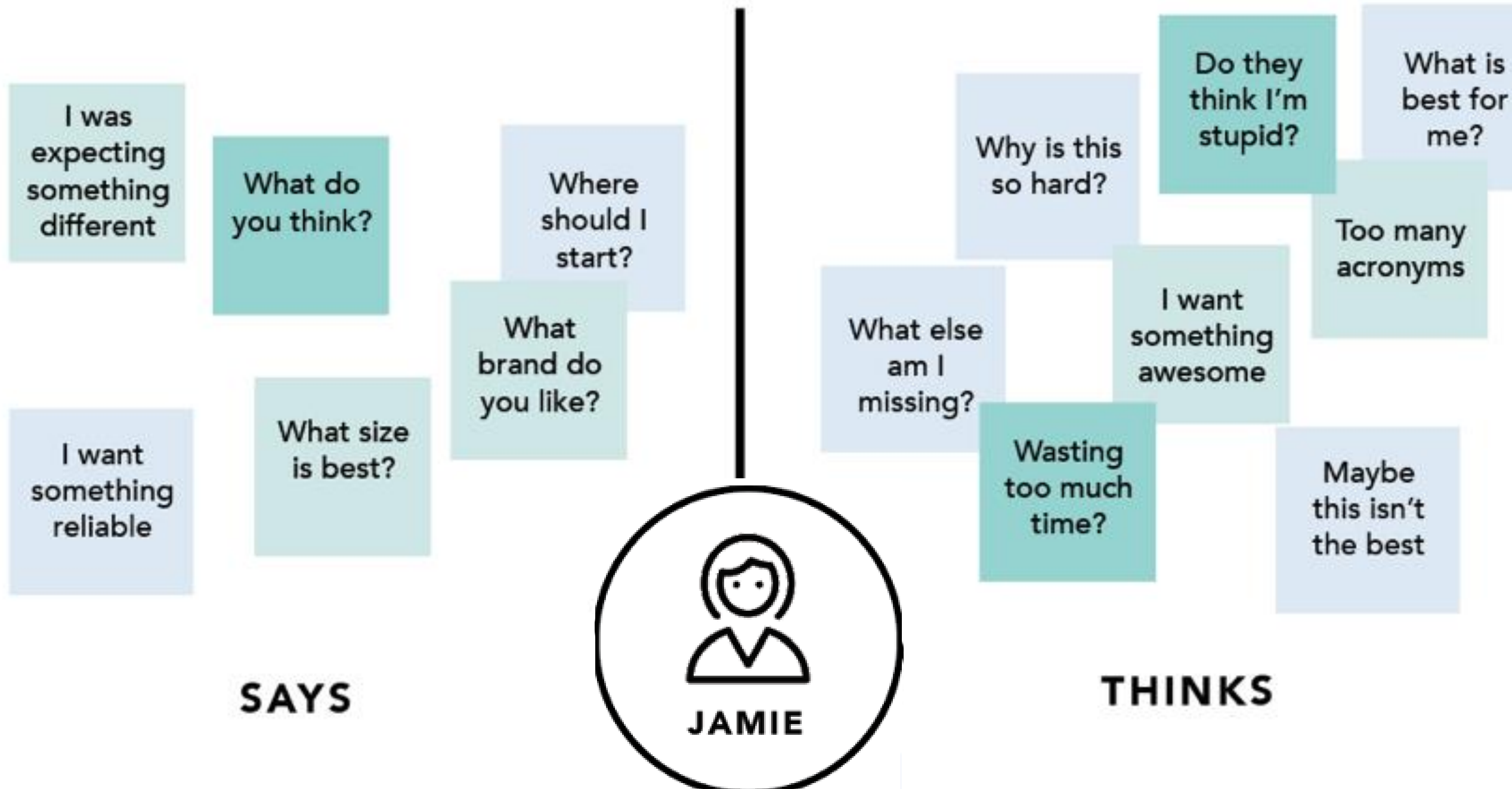
↳ WHITEBOARD OR
PRINTED OUT MAPS

↳ STICKY NOTES &
PENS / MARKERS

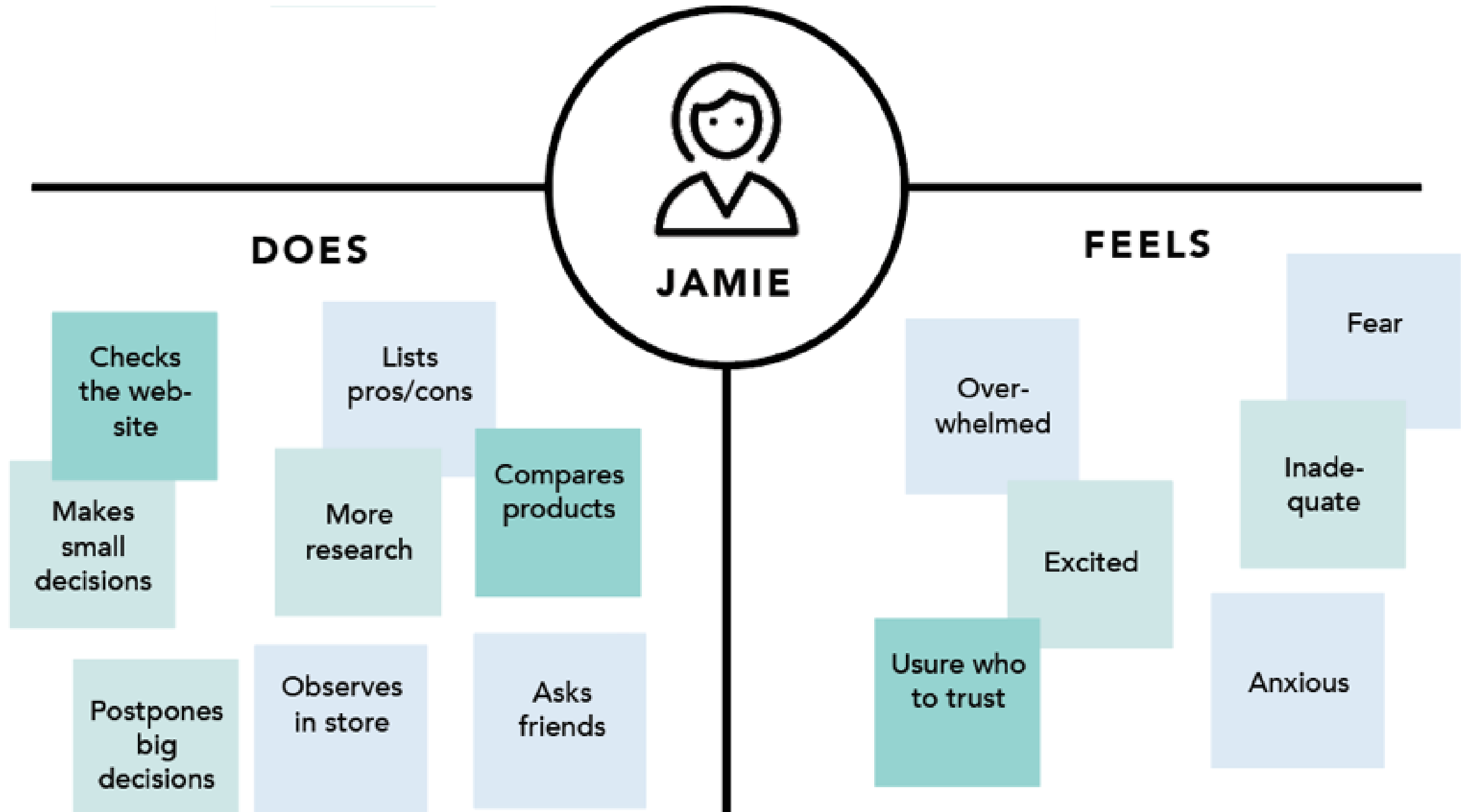


PREPARE THE DATA
YOU'VE COLLECTED
FROM THE RESEARCH
(E.G. USER INTERVIEWS
CONTEXTUAL INQUIRY
ONLINE SURVEY
TESTIMONIALS &
OTHER FEEDBACK)

Exemple Buying a TV (1/2)



Exemple Buying a TV (2/2)



2.3 Models i artefactes

2.3.5 Escenari d'ús



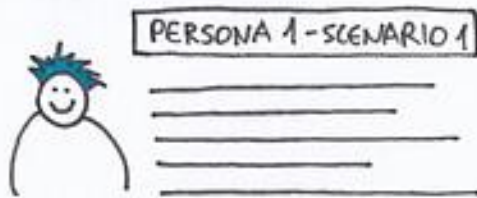
Font Flaticon

Pràctica 2

(Es defineixen 2 escenaris)

OBJECTIU: Narrar l'usuari interactuant amb el sistema per aconseguir un **objectiu** (normalment correspon amb una tasca d'usuari).

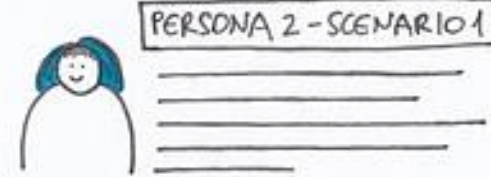
- Protagonitzat per un dels personatges ficticis.
- Té un plot: inici, desenvolupament, fi - on es descriu el context on s'inicia la necessitat d'ús, les accions i emocions de l'usuari per aconseguir el seu objectiu.
- També pot descriure situacions errònies i com l'usuari surt d'aquestes.
- La història sempre acaba bé.



PERSONA 1 - SCENARIO 1

SCENARIOS

UX Knowledge Base Sketch #16



PERSONA 2 - SCENARIO 1

SCENARIOS ARE



SHORT STORIES



SEQUENCE OF ACTIONS AND EVENTS FROM THE MAIN CHARACTER'S (PERSONA) PERSPECTIVE.



ABOUT A PERSONA



BASED ON THE PERSONA'S NEEDS, GOALS, MOTIVATIONS



TRYING TO ACHIEVE HER GOAL



BEFORE IMPLEMENTING SG, ASK: WOULD THIS BE USEFUL TO OUR USERS?



BY USING THE FUTURE PRODUCT/SERVICE



IT IS ABOUT THE FUTURE!



IN HER CONTEXT.



REAL WORLD CONTEXT CONTAINING INFORMATION ABOUT BEHAVIOR, USAGE, CONDITIONS AFFECTING THE EXPERIENCE.

SCENARIO $\neq$ USE CASE

|| SCENARIO $\neq$ USER STORY



LIKE A TECHNICAL BLUEPRINT



SHORTER (SOME SENTENCES)



IT IS ABOUT FUNCTIONS, NOT THINKING OR FEELINGS



DESCRIBING A NEED FROM A PERSONA'S PERSPECTIVE



LIST OF ACTIONS BETWEEN THE USER AND THE SYSTEM



IN THE PERSONA'S OWN WORDS.

HOW TO CREATE SCENARIOS



PREREQUISITE: PERSONAS



DECIDE WHAT SCENARIOS SHOULD BE CREATED: MOST SIGNIFICANT ACTIVITIES BY EACH PERSONA



MAKE A BELIEVABLE STORY

↳ PLOT: TRIGGER, ACTIONS, ACHIEVED GOAL

↳ CONTEXT I.E.G.: WHERE?
WEATHER CONDITIONS
LIGHT CONDITIONS
WHEN?

↳ MOTIVATIONS, REASONS

↳ HOW THE FUTURE PRODUCT/SERVICE CAN HELP REACHING THE END-GOAL?



IT IS ABOUT THE PERSONA'S EXPERIENCE (THINKING, FEELING, SEEING, DOING), NOT ABOUT UI SOLUTIONS!



TRY TO AVOID UNREALISTIC ASSUMPTIONS!
THINK ABOUT WHAT THE PERSONA WOULD DO!



EXCLUDE THE UNNECESSARY SENTENCES!



THEN YOU CAN START DEFINING REQUIREMENTS FROM THE SCENARIOS!

Exemple escenari

Títol: Notification to social security administration

“Mr. and Mrs. Macomb are retired school teachers who are now in their 70s. Their Social Security checks are an important part of their income. They've just moved to a new home and they should inform the Social Security Administration (SSA).

They don't know where the nearest Social Security office is and it's getting harder for them to do a lot of walking or driving. If it is easy and safe enough, **they would like to use the computer to notify the SSA** of their move. However, they are **nervous** about doing a task like this. They **never used computers** in their jobs. However, their son, Steve, gave them a computer last year, set it up for them, and showed them how to use **email and go to websites**. They have never been to the SSA's website, so they don't know how it is organized. Also, they are **reluctant** to give out personal information online, so they want to know how safe it is to tell the agency about their new address this way. They ...<passos sense indicar menus, clics, etc.>.....

and finally are happy for achieving communicate their new address.”

Títol: Travel agent booking a car (Hackos, 1998)

Joan Gaynor, the travel agent, takes a call from Mrs Reed. The Reed family is going on a vacation and they want Joan **to reserve a rental car for them**. When Joan talks about it he says «I'm booking a car for the Reeds. The family **includes two adults and three children**: 11, 7 and 3. They **need** a car that will hold five and all their luggage. They need a **car seat** for the toddler. They **want to pick up the car** when they fly into the airport in Salt Lake City and drop it off at the same airport a week later just before their flight back. They want to rent from an agency that has cars at the airport so they don't have to carry their luggage. They want the **cheapest rental** they can get for the week with no mileage charges.

Escenaris

Exercici: Crear Escenari (veure activitat a Material pràctiques del CV)

2.3 Models i artefactes

2.3.6 User journey map



Font Flaticon

- OBJECTIU: Representació visual que defineix l'experiència d'usuari en l'ús d'un producte.
- Protagoniste una **persona**, es representa la realització d'un **escenari**.
- Defineix les accions que realitza un usuari de forma temporal, acompanyades amb les necessitats de cada moment, els pain points (dificultats) i els “touchpoints” (o punts de contacte entre el i el producte). A més, es mostren les emocions.

Exemple Journey map

Més exemples a

<https://www.uxtweak.com/user-journey-map/examples/>

User Journey Map - Ankit Jain

Goals - Get a better experience in online train tickets booking.

Stages	→ Research & planning itinerary	→ Booking	→ Pre-Travel	→ During-Travel (Arival at platform)	→ Post-Travel
ACTIONS	- Research about destinations - Ask advice from friends & colleagues - Set a budget	- Finalize destination & date - Book tickets with prepered berth & class - Check fare & seats - Detailed information about train schedule	- Check PNR status - Check train schedules - Confirmation of seats - Packing	- Find coach and seat - Check train schedules - Look for food & other necessary options -	- Write reviews - Post photos on social media
TOUCH POINTS	- Primarily desktop research - mobile phone - Travel Agency	- IRCTC website - Check out other competitor	- Phone	- Digital bill boards - Annoucements	- Social Media websites
PAIN POINTS	- Sometimes the website takes time to load - No itinerary planner feature available - No reviews available for trains & routes - Cluttered information throughout the web platform which creates confusion - Regaining password process is frustrating	- Creating account & reseting password is painfull - During tatkal booking website server gets down - Booking modification is not available - Uncertainty in getting confirmation on 'Waiting list' tickets. - Unable to see train details as per the name	- Worried about tickets getting cancelled - Booking modification is not available	- Difficulty locating the correct platform - Can't buy travel related assesories inside the train - Tracking live updates of the train	
OPPORTUNITY	- Provide customized travel offers - Save travel history for quick booking - Remember passenger profile for autofill	- OTP based login option - Check trains based on rating system - Waiting list ticket update on a daily basis - Remember passenger profiles for autofill	- Notify for WL tickets a day before - Notify about all the important details	- Provide or suggest cross selling items for purchase like food, cab, hotel etc. - Save route details	- User can give feedback or review
EXPERIENCE	   Searching 	Frustrated 	Worried 	Confused 	Relieved 

Exemple Journey map Ankit Jain

Goal: getting a better experience in online train tickets booking.

User Journey Map - Ankit Jain

Goals - Get a better experience in online train tickets booking.

Stages	Research & planning itinerary	Booking	Pre-Trip	During Trip	Post-Trip
ACTIONS	- Research about destinations - Ask advice from friends & colleagues - Set a budget	- Finalize destination & date - Book tickets with preferred agent & class - Check train & route - Obtain information about train schedule	- Check PNR status - Check train schedule - Coordinate all details - Packing	- Post travel and enjoy - Check train schedule - Look for food & other necessary services	- Write reviews - Post photos on social media
TOUCH POINT	- Primarily desktop research - Mobile phone - Travel Agency	- IRCTC website - Check on other computer	Phone	- Digital bill boards - Advertisements	Social Media websites
PAIN POINTS	- Sometimes the website takes time to load - The website crashes before available - The website available for some & some - Cluttered information throughout the site - Confusion while entering information - Regaining password process is frustrating	- Waiting account & waiting password is long - Using hotel booking website server gets slow - Booking modification is not available - Remembering & getting confirmation on booking for tickets - People to see train details as per the route	- Checked about train getting cancelled - Booking modification is not available	- Difficulty locating the nearest platform - Getting train related information before the train - Waiting for updates of the train	
OPPORTUNITY	- Provide customized travel offers - Save travel history for quick booking - Remember passenger profile for autofill	24/7 based login option - Quick train based on rating system - Waiting for ticket update on a daily basis - Remember passenger profile for autofill	- Ready for 95% when a day before - Ready about all the important details	Provide or suggest some getting train for purchase the food, seats, hotel etc. from more details	User can give feedback or review
EXPERIENCE	 Scouting	 Frustrated	 Worried	 Confused	 Review

Stage 1: Research & Planning itinerary

- Actions

- Research about destinations
- Ask advice from friends
- Set a budget

- Touch Points

- Primarily desktop research
- Mobile phone
- Travel agency

- Pain Points

- Sometimes the website takes time to load
- No itinerary planner feature available
- No review available for trains & routes.
- Cluttered info creates confusion
- Regaining password process is frustrating

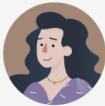
- Opportunities

- Provide customized travel offers
- Save travel history for quick booking
- Remember Passenger profile for autofill

- Experience



USER JOURNEY MAP / El Candado Inteligente Para La Bici



VALERIA
"Soy muy despistada"

ESCENARIO
Valeria va en bici a la universidad pero se le olvida poner el candado. Cuando se da cuenta, con la app activa el cierre automático y recibe confirmación. Más tarde recibe un aviso de robo, llama a seguridad y gracias a la notificación, evita el robo.

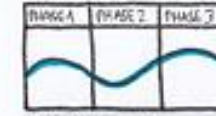
STAGES	STAGE 1 ▶	STAGE 2 ▶	STAGE 3 ▶	STAGE 4 ▶	STAGE 5
GOALS	Llegada a la universidad	Activación del cierre automático	Notificación intento de robo	Respuesta al intento de robo	Vuelve a casa
ACTIONS	- Deja la bici en el aparcamiento - Marcha hacia clase	- Llega a la clase - Recuerda que no ha puesto candado - Abre la app - Activa cierre automático	- Escucha aviso del movil - Lee el aviso	- Llama a seguridad - Sale a ver su bici - Comprueba que el candado no está dañado - Vuelve a la clase	- Sale de la clase - Va hacia el aparcamiento - Ve su bici aparcada
THOUGHTS	"Otra vez llego tarde a clase"	"Menos mal que me he acordado y puedo cerrar desde aquí"	"Ostras, me estan robando, necesito hacer algo "	"Confío en que seguridad o yo lleguemos a tiempo"	"Menos mal que el candado me ha avisado"
PAIN POINTS	Olvido	Posible fallo de la app o del candado	Interrumpe la clase con el ruido	Pierde parte de la clase Está estresada	
EMOTIONS	 Contenta	 Aliviada	 Ansiedad	 Estresada	 Happy
TOUCHPOINTS					
OPPORTUNITIES	La usuaria recibe recordatorio de cierre cuando llega a la universidad.	Si falla la app, la usuaria puede recibir aviso por otros canales, ej. el reloj inteligente.	La usuaria puede configurar que avise con vibración. El aviso indica coordenadas de intento de robo.	La usuaria recibe un aviso si se ha dañado el candado. Colaboración con seguridad de la uni.	



INTRODUCTION TO JOURNEY MAPPING PART 1

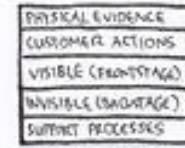
UX Knowledge Base Sketch #54

SIMILAR DIAGRAMS:



EXPERIENCE MAP

↳ BROADER CONTEXT
OF HUMAN BEHAVIOR
↳ BUSINESS-/PRODUCT
AGNOSTIC



SERVICE BLUEPRINT

↳ FOCUSES ON HOW
THE SERVICE IS PROVIDED:
ROLES, PROCESSES -
EVEN BACKSTAGE!



DEFINITION

CHRONOLOGICAL: EXPERIENCE OVER TIME
WITH A PRODUCT/SERVICE
IN ORDER TO ACHIEVE A GOAL
VISUALIZING A SEQUENCE OF STEPS & TOUCHPOINTS
FROM THE USER'S/CUSTOMER'S PERSPECTIVE

DESIGNER:
UX CONCLUSIONSDEVELOPER:
TECHNOLOGICAL
CONCLUSIONSPRODUCT MANAGER:
BUSINESS CONCLUSIONS

COLLABORATIVE

~ COMMON LANGUAGE
⇒ SHARED UNDERSTANDING
• MULTIDISCIPLINARY APPROACH!
• IDEAL FOR CROSS-FUNCTIONAL TEAMS
• INCREASES ENGAGEMENT
(THE MAP SERVES AS A SO-CALLED
BOUNDARY OBJECT)

(NOT ONLY THESE 3 ROLES! E.G. UX WRITER: "WHAT QUESTIONS SHOULD
BE ANSWERED AT EACH STEP?")

FLEXIBLE &
VERSATILE
TOOL

ANY FORMAT THAT
SUPPORTS THE GOAL!



INCLUDES

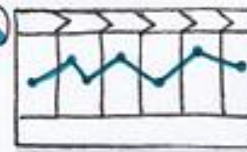
TOUCHPOINTS: STEPS THAT REPRESENT INTERACTIONS
WITH THE BRAND.
E.G.:
↳ EMAIL
↳ PURCHASE
↳ READING TESTIMONIALS
↳ REVIEWS (E.G. ON A 3RD PARTY-SITE)
↳ EXTERNAL AS WELL!

SOME TYPICAL
GOALS:

- UNDERSTANDING NEEDS,
MOTIVATIONS
- LOOKING FOR GAPS,
OPPORTUNITIES
- IDENTIFY PAIN POINTS
- HELPING COLLABORATION
(SHARED VISION)
- BUILDING EMPATHY

ELEMENTS:

- NOT A DEFINITIVE LIST! CREATE A COMBINATION (ADD
CUSTOM LANES) TO BEST SERVE THE PURPOSE/GOAL!



1. TIMELINE →
 2. PHASES → STAGES OF THE EXPERIENCE
 3. WHO (PERSONA) → (IT CAN INCLUDE MULTIPLE: COMPARATIVE)
 4. SCENARIO →
 5. STEPS - IN A SEQUENCE → INTERACTIONS - E.G. ACTIVITIES WAITING
 6. CHANNELS → FOR VISUALIZING MULTI-CHANNEL JOURNEYS
EXAMPLES: IN-PERSON, WEB, PRINT, APP → METHODS OF INTERACTING
 7. THOUGHTS → E.G. - QUOTES FROM RESEARCH DATA EXPECTATIONS ETC.
 8. EMOTIONAL EXPERIENCE → LEVEL OF SATISFACTION THROUGHOUT THE JOURNEY
 9. DRAMATIC ARC → LEVEL OF ENGAGEMENT
 10. STAKEHOLDERS/OWNERSHIP → WHO IS INVOLVED / RESPONSIBLE
(IF YOU ALSO INCLUDE BACKSTAGE PROCESSES: SERVICE BLUEPRINT)
 11. INSIGHTS: PAINPOINTS, OPPORTUNITIES RISKS, THREATS (WHAT CAN GO WRONG?)
- ooo YOU CAN ADD MANY MORE THINGS!
E.G. STORYBOARDS, BARRIERS (SEE PART 2!)

IT CREATES A
STRUCTURED,
VISUALIZED
NARRATIVE,
A STORY:

WHAT THE MAIN
CHARACTER DOES,
THINKS, FEELS -
WHILE ACHIEVING
A GOAL & HAVING
EXPERIENCE WITH
THE PRODUCT/SERVICE

MAPPED OUT ON A
TIMELINE.

CONTINGUT

2.1 Introducció

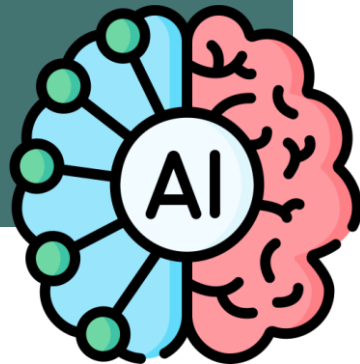
2.2 Tècniques d'investigació d'usuaris

2.3 Models i artefactes

2.4 Investigació d'usuaris en productes amb IA

2.5 IAG en la Conceptualització

2.4 Investigació d'usuaris en productes amb IA



Dissenyar experiències d'usuari amb IA no és només integrar l'IA en la interfície, és validar si el problema (que es vol solucionar amb l'IA) realment existeix i dissenyar amb incertesa.

2.4 Investigació d'usuaris en productes amb IA



Yildirim, N., Pushkarna, M., Goyal, N., Wattenberg, M., & Viégas, F. (2023, April). Investigating how practitioners use human-ai guidelines: A case study on the people+ ai guidebook. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (pp. 1-13).

[people+ ai guidebook \(PAIR\)](#) de google

Conclusions rellevants:

- Els productes amb IA fallen quan no responen a una necessitat humana real.
- En la guia PAIR (i similars) falta suport en l'etapa temprana “problem framing”
- Es produeix un procés híbrid que intenta reconciliar dos aproximacions “user-centric” i “AI-centric”.

2.4 Investigació d'usuaris en productes amb IA



La recerca d'usuaris serveix per:

Identificar moments de:



- Carga cognitiva alta
- Incertesa
- Repetició
- Decisió complexa

...

Saber més sobre relació
usuari-IA (confiança,
expectatives, ...)

Per tant, tot això s'ha de
tenir en compte quan
dissenyem entrevistes,
qüestionaris, etc.

2.4 Investigació d'usuaris en productes amb IA



La recerca d'usuaris serveix per:

Identificar moments de:



- Carga cognitiva alta
- Incertesa
- Repetició
- Decisió complexa

...

Saber més sobre relació
usuari-IA (confiança,
expectatives, ...)



Preguntar-nos:

- Aquí l'IA redueix l'esforç?
- Aquí l'IA millora la claredat?
- Aquí l'IA millora la presa de decisions?

...



- Té sentit usar l'IA?
- Quin tipus?
- Amb quin nivell d'automatització i control de l'usuari
- ...

2.4 Investigació d'usuaris en productes amb IA



Liao, Q. V., Subramonyam, H., Wang, J., & Wortman Vaughan, J. (2023, April). **Designerly understanding: Information needs for model transparency to support design ideation for AI-powered user experience.** In *Proceedings of the 2023 CHI conference on human factors in computing systems* (pp. 1-21).

“Designerly understanding”: **suficient comprensió de l'IA per poder-la utilitzar** com a material de disseny, sense necessitat d'entendre'n la part interna.

Conclusions rellevants:

- El dissenyadors necessiten comprendre capacitats, limitacions i riscos
- El dissenyadors creen regles (tècniques, contextuais, ètiques) per utilitzar l'IA ex. “mostrar avís si la “confidence” és baixa”
- Dissenyar amb IA implica establir criteris, gestionar **incertesa** i anticipar riscos per l'usuari (responsible design)

CONTINGUT

2.1 Introducció

2.2 Tècniques d'investigació d'usuaris

2.3 Models i artefactes

2.4 Investigació d'usuaris en productes amb IA

2.5 IAG en la Conceptualització

2.5 IAG en la conceptualització

La IA generativa MAI
no ha de substituir als
usuaris reals!

2.5 IAG en la conceptualització

Russell, D. M. (2025). The Challenges of Synthetic Users in UX Research. Interactions, 33(1), 7-8.

Reptes d'usuaris sintètics:

- Comportaments previsibles
- Risc de biaxos
- Limitacions dades qualitatives (observació al context real)
- Manca d'imprevisibilitat humana (emocions)

La IA generativa MAI
no ha de substituir als
usuaris reals!

2.5 IAG en la conceptualització

QUAN

- Abans del user research
- Després del user research

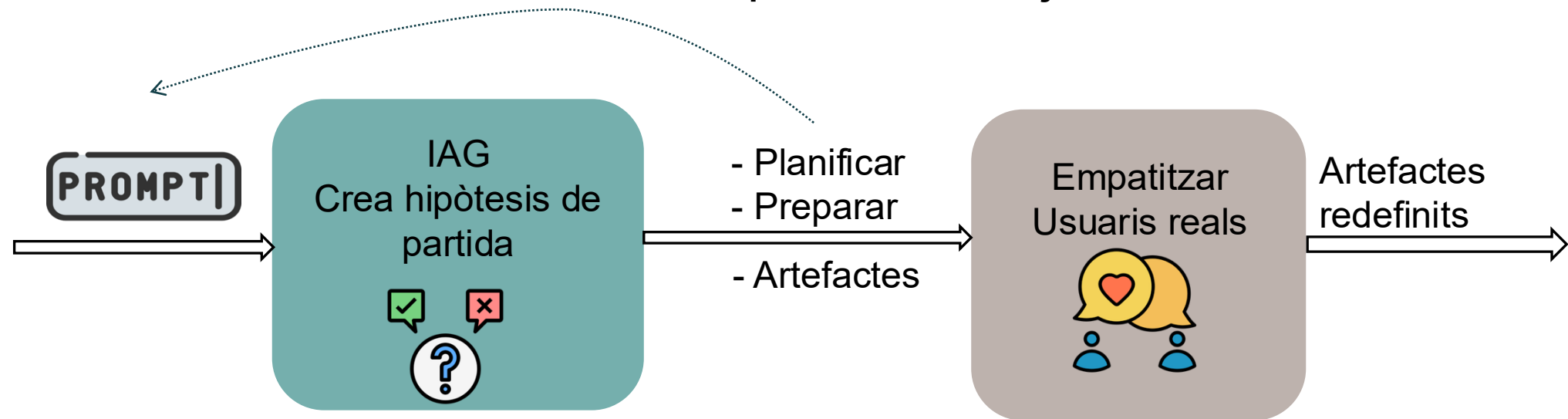
COM

- **Prompt engineering**: és la disciplina que estudia com dissenyar, estructurar i optimitzar instruccions (prompts) per obtenir respostes més precises, útils i consistents d'un model de llenguatge. <https://www.promptingguide.ai/introduction>

2.5 IAG en la conceptualització

ABANS del User Research

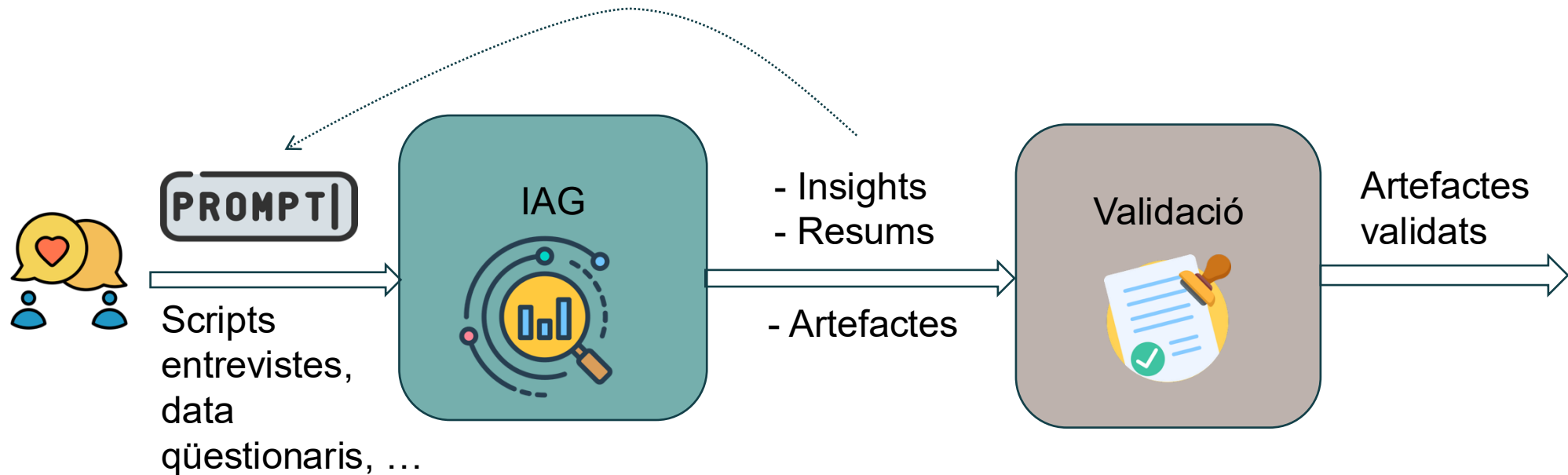
Per a planificar, preparar i per obtenir hipòtesis de partida sobre els nostres usuaris (ex. Proto-persona). Després empatitzem amb els usuaris reals, validem les hipòtesis i re-ajustem artefactes.



2.5 IAG en la conceptualització

DESPRÉS del User Research

Per analitzar les dades i extreure insights (temes comuns, pain points...) i definir artefactes (Persona, escenaris, etc.) tenint com base les dades recollides.



2.5 IAG en la conceptualització

Prompts per:

- Planificar user research.
- Recollir dades (preparar entrevistes, qüestionaris, ...)
- Anàlisi de dades (privacitat: anonimitza dades d'usuaris reals).
- Comunicar resultats.

<https://www.userinterviews.com/blog/ai-prompts-for-ux-research>

<https://www.nngroup.com/articles/research-with-ai/>

Exemple de prompt per la planificació

<https://www.looppanel.com/blog/14-chat-gpt-prompts-to-use-for-ux-research> - Prompts

I need to do a UX research project for my product X, a [describe the product]. The research goal is to find out [describe the goal]. The target audience is [add user demographics, numbers, and relevant details]. What research methods would be right for this project? Give me a broad research plan for each option.

2.5 IAG en la conceptualització

Exemple en 3 etapes: creació Persona i journey map, app for exercise routines

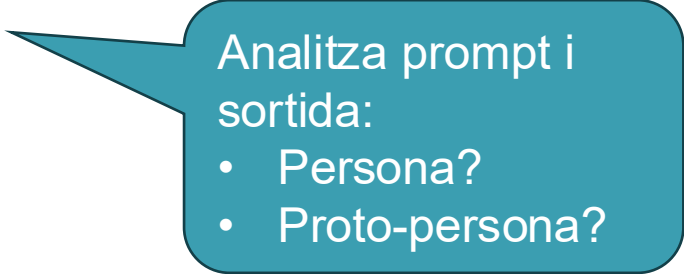
Font <https://medium.com/design-bootcamp/revolutionizing-personas-unleashing-the-power-of-ai-in-ux-design-4f66010eb1f8>

Step 1/3. Identify the target market for the problem statement

Grounding the personas on a specific problem

Problem statement: “Many fitness enthusiasts and professionals struggle to find engaging and effective workouts that maintain a consistent tempo and rhythm. This can lead to decreased motivation, difficulty in following exercise routines, and suboptimal results”.

Prompt: “What personas are impacted by the following problem statement? [problem statement]”



Analitza prompt i sortida:

- Persona?
- Proto-persona?

2.5 IAG en la conceptualització

Exemple en 3 etapes: creació Persona i journey map, app for exercise routines

Font <https://medium.com/design-bootcamp/revolutionizing-personas-unleashing-the-power-of-ai-in-ux-design-4f66010eb1f8>

Step 2/3. Create your Persona

Choose your primary persona or ask GPT to pick it for you

Prompt: “Create a UX Persona for **[primary persona]**. Give it a unique name. Include Pain points, Goals, Motivations, Demographic, Empathy map, Psychographics, Buyer and Purchasing influences, Online behaviour, Interests, Technology proficiency, Typical hobbies, and any additional relevant research to help inform decisions. Please provide supporting references.”

Analitza prompt i sortida:

- Persona?
- Proto-persona?

2.5 IAG en la conceptualització

Exemple en 3 etapes: creació Persona i journey map, app for exercise routines

Font <https://medium.com/design-bootcamp/revolutionizing-personas-unleashing-the-power-of-ai-in-ux-design-4f66010eb1f8>

Step 3/3. Describe their current journey

Note: This is a hypothesis, which means this is not intended to skip your own research. Use this as a starting point.

Prompt: *“Describe [persona]’s current experience (no solution) in a UX journey map. Tabulate the results. Provide supporting research references.”*

Step 4. Bring your Persona to life with images

Use AI generative (LeonardoAI, DALL-E 3...)

○ <https://this-person-does-not-exist.com/es>

2.5 IAG en la conceptualització

[Persona creation AI prompts \(.pdf\)](#)

Analitza prompt i resultat:

- Persona?
- Proto-persona?

Code Your Research Data, Step-By-Step

AI can take some of the grunt work out of research organization. Upload your research data in spreadsheet format and start with this prompt:

You are an expert UX researcher.

Review the attached research data. Identify [observations/quotes/notes] that share similarities, and group them into broad, abstract categories. Assign each [observation/quote/note] to one of these categories in a new column labeled "Group Level 1" in the provided .xlsx file. Clearly name each category to reflect its content. After assigning categories, sort the spreadsheet alphabetically by the "Group Level 1" column.

Once you have your preliminary groups, you can use the following prompt to narrow down and get more specific:

2.5 IAG en la conceptualització

When You Don't Want to Use ChatGPT for UX Design?



-  Limited Context Understanding
-  Hallucination
-  Ethical Concerns
-  Over-Reliance on Automation
-  Limited Interdisciplinary Knowledge

Interaction Design Foundation
interaction-design.org

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- <https://medium.com/design-bootcamp/revolutionizing-personas-unleashing-the-power-of-ai-in-ux-design-4f66010eb1f8>

